images of nuclear mushroom clouds juxtapose the explosion of atomic bombs over Hiroshima and Nagasaki with the testing of thermonuclear weapons. Yet the advent of thermonuclear weapons during the 1950s, and the seemingly unfettered build-up of nuclear arsenals, posed a new and existential threat to humankind (and indeed all organic life). In particular, thermonuclear weapons were able to cause huge amounts of lethal radioactive fallout. The amount of fallout depended on whether the weapons were ground-burst or air-burst (depending primarily on the target). The ground-bursts would create the greater amount of fallout.

Immediate casualties in a full-scale thermonuclear war would likely have been measured in hundreds of millions. Each B-52 bomber carried either two Mark 39 bombs (each with a yield of four megatons) or four Mark 28 bombs (each with a yield of two megatons), together with a Hound Dog inertially guided cruise missile, with a four-megaton warhead.[20] The British also developed a thermonuclear bomb, Yellow Sun Mark-2, with a megaton yield. The Thor and Jupiter IRBMs (as well as the Atlas ICBM) were armed with the W49 1.44 MT warhead. The largest thermonuclear weapons in the US arsenal were the Mark 17 and Mark 24 bombs with yields of between fifteen and twenty megatons.[21] As noted, the Soviets tested the fifty-eight megaton Tsar Bomba in 1961, though it proved too large to be deployed as a weapon. Soviet ICBMs, IRBMs, MRBMs, and long-range bombers, could all be equipped with megaton-yield weapons.

What if the United States had mounted ‘a full retaliatory response’ against the Soviet Union? According to Daniel Ellsberg in his 2017 book, *Doomsday Machine: Confessions of a Nuclear War Planner,* the Joint Chiefs of Staff informed the President in 1961 that if the US nuclear war plan (SIOP-62) was implemented, an estimated 275 million people would die in the USSR and the PRC, rising to 325 million people within six months.[22] Ellsberg was a RAND corporation analyst, working as a consultant in the Pentagon in 1961, and discovered aspects of US nuclear command and control unknown to Kennedy’s advisers. Ellsberg uncovered arrangements for delegating authority to the military to use nuclear weapons, which were then unknown to the President or his senior civilian officials.